Lab Program 2

Develop a multistage dockerfile for container Orchestration

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Step 1 : Directory and file creation

Create the all the necessary files as shown below

```
❯ tree -I node_modules
.
├── dist
├── Dockerfile
├── package.json
├── package-lock.json
└── src
    └── index.js

3 directories, 4 files
❯ ls
dist  Dockerfile  node_modules  package.json  package-lock.json  src
```

Step 2 : Add necessary content

```
//src/index.js
const express = require("express")
const app = express()
const PORT = 3000

app.get('/', (req, res) => {
res.send('Hello from multi stage docker ')
})

app.listen(PORT, () => {
console.log(`server running on port ${PORT}`);
})
```

Add build script in package.json

```
//package.json
{
"name": "l3",
"version": "1.0.0",
"description": "",
"main": "index.js",
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1",
"build": "mkdir -p dist && cp -r src/* dist/"
},
"keywords": [],
"author": "",
"license": "ISC",
"dependencies": {
"express": "^5.1.0"
}
}
```

Step 3 : Create Dockerfile

```
#Dockerfile
FROM node:20-alpine AS builder

WORKDIR /app
```

```
COPY package.json package-lock.json ./

RUN npm install

COPY . .

RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/package.json ./

COPY --from=builder /app/package-lock.json ./

COPY --from=builder /app/dist ./dist

COPY --from=builder /app/node_modules ./node_modules

EXPOSE 3000

CMD [ "node","dist/index.js" ]
```

Step 4 : Build an image

```
> docker build -t secondprog .
```

```
❯ docker build -t secondprog .
[+] Building 2.4s (16/16) FINISHED                                  docker:default
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 427B                                          0.0s
 => [internal] load metadata for docker.io/library/node:20-alpine             2.3s
 => [auth] library/node:pull token for registry-1.docker.io                   0.0s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df28  0.0s
 => [internal] load build context                                             0.0s
 => => transferring context: 43.51kB                                          0.0s
 => CACHED [builder 2/6] WORKDIR /app                                         0.0s
 => CACHED [builder 3/6] COPY package.json package-lock.json ./               0.0s
 => CACHED [builder 4/6] RUN npm install                                      0.0s
 => CACHED [builder 5/6] COPY .  .                                            0.0s
 => CACHED [builder 6/6] RUN npm run build                                    0.0s
 => CACHED [stage-1 3/6] COPY --from=builder /app/package.json ./             0.0s
 => CACHED [stage-1 4/6] COPY --from=builder /app/package-lock.json ./        0.0s
 => CACHED [stage-1 5/6] COPY --from=builder /app/dist ./dist                 0.0s
 => CACHED [stage-1 6/6] COPY --from=builder /app/node_modules ./node_modules  0.0s
 => exporting to image                                                        0.0s
 => => exporting layers                                                       0.0s
 => => writing image sha256:457acbbee23b3bd563c2cd57b32dfa056eaaea448417bf9144e913688  0.0s
 => => naming to docker.io/library/secondprog                                 0.0s
```

```
❯ docker images | grep secondprog
secondprog                latest        457acbbee23b   32 minutes ago   137MB
```

Step 5: Run and verify

```
Run 'docker run --help' for more information
❯ docker run -p 5500:3000 secondprog
server running on port 3000
```

```
❯ curl -X GET localhost:3000
curl: (56) Recv failure: Connection reset by peer
❯ curl -X GET localhost:5500
Hello from multi stage docker
```